

Roll No. _____

B.Tech IV sem (Main/Back) Exam April-May 2026**Artificial Intelligence & Data Science****4AID2-01 Discrete Mathematics Structure****Paper Code : 4E1301****CS, IT, AID, CAI, CCS, CDS, CIT, CSR****Time: 3 Hours****Maximum Marks: 70**

Attempt all ten questions from Part A, five questions out of seven questions from Part B and three questions out of five questions from Part C.

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. Units of quantities used/ calculated must be stated clearly.

Use of following supporting material is permitted during examination. (Mentioned in form No.205)

1. _____

2. _____

Part A (Answer should be given up to 25 words only)
All questions are compulsory

- Q.1 Define countable set.
 Q.2 Define into and onto function.
 Q.3 What do you mean by "conjunction" connection ($p \wedge q$).
 Q.4 Define finite state machine.
 Q.5 Define well ordered set.
 Q.6 From 6 boys and 4 girls, 5 are to be selected for admission for a particular course. In how many ways can this be done if there must be exactly 2 girls.
 Q.7 Define abelian group.
 Q.8 Define Homomorphism of group.
 Q.9 Define planer graph.
 Q.10 Define chromatic number.

10 x 2 = 20

Part B (Analytical/Problem solving questions)
Attempt any Five questions

- Q.1 The relation R on the set S of all real numbers is define as : $a R b$ is and only if $1+ab > 0$. Show that this relation is reflexive and symmetric not transitive.
 Q.2 In a group of 52 person, 16 drink tea but not coffee and 33 drink tea. Find
 i. How many drink tea and coffee both
 ii. How many drink coffee but not tea.
 Q.3 Prove that $(p \Rightarrow q) \vee r \equiv (p \vee r) \Rightarrow (q \vee r)$
 Q.4 Let $A = \{2, 3, 4, 6, 12, 36, 48\}$ be a non empty set and R be the partial order relation on divisibility defined on A, i.e. if $a, b \in A$ then a divides b.
 Draw the graph and Hasse diagram.

- Q.5 Explain bounded and complemented lattices.
Q.6 Prove that every cyclic group is abelian.
Q.7 Explain Eulerian and Hamiltonian graph.

5x 4 = 20

Part C(Descriptive/Analytical/Problem Solving/Design question)

Attempt any three questions

- Q.1 Prove by the principle of mathematical induction :

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

- Q.2 Let M be a finite state machine with state table given below :

State	f (input)		g (output)	
	a	b	a	b
S ₀	S ₁	S ₂	x	x
S ₁	S ₀	S ₁	x	y
S ₂	S ₁	S ₂	y	x
S ₃	S ₁	S ₂	y	y

- (i) Find the input set I, the state set S, the output set o and the initial state of M.
(ii) Draw the state diagram of M.

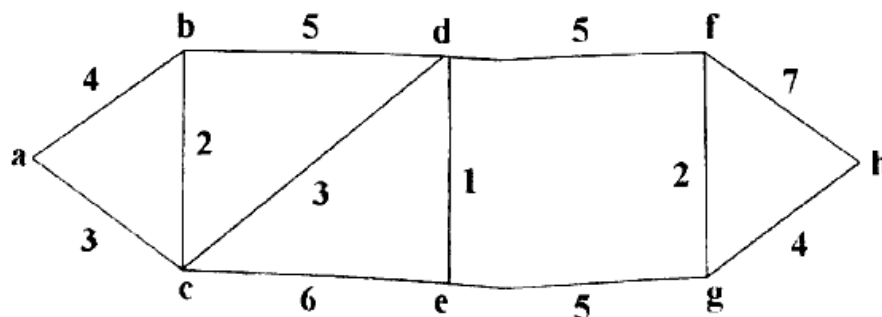
- Q.3 Solve the recurrence relation.

$$a_n - 8a_{n-1} + 16a_{n-2} = 0$$

with initial condition $a_2=16, a_3=80$.

- Q.4 Show that set of all square matrix of order (mxn) under the binary operations addition and multiplication is a non commutative ring.

- Q.5 Find the shortest path and its length between the vertices a and h in the following weighted graph:



3 x 10 = 30